

General Observations

In vector a, the 4th dimension (w) was never transformed, since its value was 0. This shows that vectors representing the direction can't be moved. Only rotated, scaled, or reflected.

Vector b was always moved relatively to the 4th dimension, because w was non-zero

Task 1)

$\text{Det}(A) = 1$ keeps everything in place, and the matrix values perform only movement in space.

Task 2)

$\text{Det}(A) = -1$ shows that the matrix mirrors objects. In the matrix, we see -1, which flips the direction of the axis.

Task 3)

$\text{Det}(A) = 1$ with $\sqrt{2}/2$ shows that the matrix is purely rotational, without volume and dimension change.

Task 4)

The matrix changes the vector scale, and $\text{Det}(A) = 0.75$ shows that the new vector keeps 75% of the original "volume"